1. **ATM Machine Controller (Verilog FSM)**

   One evening, while walking along GS Road, we stopped at an ATM to withdraw money. The machine should securely handle different user needs, modeled as a Finite State Machine (FSM) in Verilog HDL. The design must consider both transaction flows and PIN management flows, ensuring smooth, secure, and real-life functioning. Model the controller as an FSM and provide a testbench that demonstrates correctness across realistic scenarios figure out the number of LUTs, flip flops, power, latency, maximum operating frequency, and delay required in the corresponding designs.

   Scenario 1: Cash Withdrawal

   1. Insert ATM card.
   2. Enter PIN. (If wrong 3 times → card blocked).
   3. Select Withdraw option.
   4. ATM asks: Savings or Current account.
   5. Enter Withdrawal Amount.
   6. ATM checks two conditions:
      - Account has sufficient balance.
      - ATM has enough cash bins to meet that amount.
   7. The ATM should dynamically display only those denominations (Rs. 100, Rs. 500, Rs. 2000) that are available at that moment and possible for the entered amount. Example: If Rs. 500 bin is empty, only Rs. 100 and Rs. 2000 appear as selectable.
   8. After selecting denomination, ATM dispenses cash, prints receipt (if requested), updates balance, and ejects card.
   9. If any condition fails (insufficient balance, no matching denominations, or Cancel pressed) → transaction terminates safely, card is returned, and message displayed.

   Scenario 2: New PIN Set (Fresh Card)

   1. Insert card → ATM detects that no PIN is set yet.
   2. System asks user to enter last 4 digits of account number.
   3. OTP is sent to registered mobile number.
   4. User enters OTP → if correct, ATM allows them to set a new PIN.
   5. Confirmation message displayed and card ejected.

   Scenario 3: PIN Change / Reset (Existing Card)

1. Insert card → enter current PIN.
2. From menu, select Change/Reset PIN.
3. System asks for last 4 digits of account number.
4. OTP sent → on entering correct OTP, ATM prompts for new PIN.
5. New PIN confirmed → message displayed, optional receipt printed, card ejected.

Additional Scenarios

- **Mini Statement:** PIN verified → select option → last 5 transactions printed.
- **Balance Inquiry:** PIN verified → balance displayed, optional receipt printed.
- **Cancel Option:** At any stage, Cancel button → aborts transaction → eject card safely.

FSM Requirements

- States include: `IDLE`, `CARD_INSERTED`, `PIN_ENTRY`, `ACCOUNT_VERIFY`, `TRANSACTION_MENU`, `WITHDRAW`, `ACCOUNT_TYPE_SELECT`, `AMOUNT_ENTRY`, `DENOMINATION_SELECT`, `CHECK_BALANCE`, `DISPENSE_CASH`, `RECEIPT_OPTION`, `BALANCE_INQUIRY`, `MINI_STATEMENT`, `NEW_PIN_SETUP`, `PIN_RESET`, `OTP_VERIFY`, `PIN_CONFIRM`, `CARD_EJECT`, `COMPLETE`, `CANCELLED`, `BLOCKED`, `ERROR`.
- Must handle:
  1. Wrong PIN attempts (max 3).
  2. Last 4 digits + OTP for new PIN set/reset.
  3. ATM cash bin availability check for denominations.
  4. Cancel button & hardware errors.

2. Design a sequence detector in Verilog HDL to detect the bit pattern **1011** in a serial input stream.

- Implement the detector using both **Moore** and **Mealy** FSM models.
- The detector must support **overlapping sequences**.
- For input stream **1011011011000110101101**, the output should count the number of detection.

Tasks:

1. Develop the state diagram for both Moore and Mealy FSM implementations.
2. Construct the corresponding state transition tables.
3. Write Verilog HDL code for both FSM models.
4. Verify functionality using a testbench with multiple input streams, including cases with overlapping occurrences of the sequence and compare the number of LUTs, flip flops, power, latency, maximum operating frequency, and delay required in the corresponding designs.